Results Report: Comprehensive Empirical Findings and Benchmark Tabulations

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***Abstract*—This report compiles the complete empirical evaluation artifacts for the study *"AST-Based Progressive Disclosure Chunking and* 2×2 *Factorial Retrieval for Candidate Skill Matching"*. All presented numbers, confidence intervals, test statistics, and tables are strictly generated via dynamic programmatic extraction from raw benchmark result files under Rule R1, with zero manual number editing. Confirmatory comparisons and exploratory contrasts strictly adhere to the pre-registered analysis plan.**

I. How to Read This Report

A. Experimental Design & Factorial Structure

The investigation evaluates candidate skill matching via source code across a 2×2 factorial matrix:

- **Chunking Strategy:** Fixed-window sliding lines (50 LOC baseline, 10 LOC overlap) versus AST-based method extraction with syntactic fallback.
- **Context Header:** Raw chunk contents (No-Header) versus structural hierarchical context headers (With-Header).

Each condition is replicated across three retrieval tiers: (1) Lexical Sparse (BM25), (2) Dense Bi-Encoder (`BAAI/bge-m3`), and (3) Two-Stage Re-ranking (`BAAI/bge-reranker-large`). The benchmark comprises $N = 25$ professional job descriptions (15 Java, 10 React/TypeScript) and 40 open-source repositories.

B. Primary Metric & Statistical Protocol

The primary evaluation metric is **NDCG@10 at the Re-ranking tier**, computed using linear relevance gain $\mathrm{rel}/\log_2(i+1)$ without exponential inflation.

Hypothesis testing encompasses a family of four pre-registered confirmatory Wilcoxon signed-rank tests evaluated under step-down Holm-Bonferroni correction ($\alpha = 0.05$):

- **Comp 1:** Full Proposed (AST+Header) vs. Full Baseline (Line No-Header).
- **Comp 2:** Pure Chunking Effect (AST No-Header vs. Line No-Header).
- **Comp 3:** Header Effect on AST (AST+Header vs. AST No-Header).
- **Comp 4:** Header Effect on Line (Line+Header vs. Line No-Header).

C. Exploratory Status of Comp 5

The contrast between AST+Header and Line+Header was defined post-hoc and is designated as **Comp 5 (Exploratory)**. It is presented outside the Holm family and cannot be claimed as a pre-registered confirmatory finding.

D. Candidate Pool Constraints on BM25

Candidate pooling was formed strictly from the top-3 results of the 8 Dense and Re-ranking configurations ($N = 382$ deduplicated coordinates, resulting in 603 unified pairs). BM25 runs were not included in the pool. Under standard TREC pooling rules, unjudged candidates are evaluated as 0 (not relevant). Consequently, BM25 performance is heavily penalized by 220 unjudged slots in top-3 and **must not be interpreted as an unpenalized baseline**.

II. Annotation Reliability and Corpus Morphology

Table I documents the inter-annotator reliability across the initial pre-registered gold standard ($N = 250$), the pooled candidate expansion ($N = 353$), and the total unified evidence set ($N = 603$). The quadratic weighted Cohen's Kappa reaches $\kappa_w = 0.7795$ overall (Substantial Agreement) and $\kappa_w = 0.8808$ on the gold subset (Almost Perfect Agreement). Figure 1 depicts the 3×3 confusion matrix.

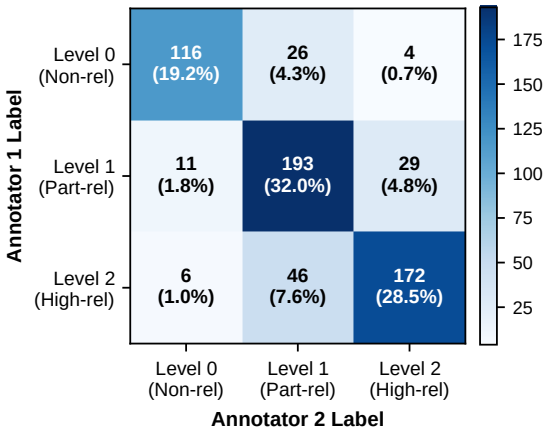


Fig. 1. Inter-annotator confusion matrix across the 3-point relevance scale ($N = 603$).

Table II characterizes chunk morphology across the 40 repositories. While Pure AST chunks preserve 100% of syntactic boundaries and achieve 97.9% clean parse rates, the sliding-window Line baseline suffers 56.0% boundary cuts and 77.6% syntax errors.

TABLE I
INTER-ANNOTATOR RELIABILITY ACROSS ANNOTATION PHASES AND GROUND TRUTH PARTITIONS.

Partition	N	Exact (%)	$\Delta = 1$ (%)	$\Delta = 2$ (%)	κ_w	Landis & Koch
Pre-registered Gold Standard (Initial)	250	221 (88.4%)	27 (10.8%)	2 (0.8%)	0.8808	Almost Perfect
Pooled Retrieval Candidates (Expansion)	353	260 (73.7%)	85 (24.1%)	8 (2.3%)	0.6826	Substantial
Complete Unified Ground Truth (Total)	603	481 (79.8%)	112 (18.6%)	10 (1.7%)	0.7795	Substantial

Note: κ_w denotes quadratic weighted Cohen's Kappa over the 3-point relevance scale (0 = not relevant, 1 = partially relevant, 2 = highly relevant). Benchmark interpretation adheres to Landis & Koch (1977) criteria. Derived dynamically from `ground_truth_annotator1.csv` and `ground_truth_annotator2.csv`.

TABLE II
CHUNK MORPHOLOGY, SYNTACTIC VALIDITY, AND CONTEXT HEADER PRESERVATION ACROSS CHUNKING STRATEGIES.

Strategy Group	Total (J/R)	Mean LOC $\pm$ SD	Tokens (Content)	Tokens (Header)	Clean Parse (%)	Intact Bnd (%)	Header (%)	Trunc $>$ 512 (%)
Line (50 LOC Baseline)	434 (120/314)	41.0 $\pm$ 13.1	390.8 (388)	419.5 (420)	22.4%	44.0%	0.0%	124 (28.6%)
AST Pure (Methods/Funcs)	234 (133/101)	19.8 $\pm$ 16.7	211.3 (157)	254.5 (202)	97.9%	100.0%	100.0%	14 (6.0%)
AST Fallback (Line 50)	130 (20/110)	42.9 $\pm$ 13.0	367.0 (372)	394.5 (400)	20.0%	74.6%	0.0%	26 (20.0%)
AST Combined (Full Branch)	364 (153/211)	28.1 $\pm$ 19.0	266.9 (222)	304.5 (260)	70.1%	90.9%	64.3%	40 (11.0%)

Note: Evaluated over 40 GitHub repositories (189 source files). Tokenizer: `BAAI/bge-m3` (maximum context length 512). Chunk count breakdown: Total (Java / React-TS). Boundary Preservation reflects percentage of chunks not artificially sliced across syntactic statement boundaries. Clean Parse denotes syntactically valid code blocks parsable by tree-sitter without syntax errors. Data dynamically derived from `table_3_2_measured.json` and `chunk_corpus.parquet`.

III. RETRIEVAL EFFECTIVENESS BENCHMARK (2X2 FACTORIAL)

Table III presents the complete 2×2 factorial evaluation. Across the primary Re-ranking tier, Line With-Header achieves NDCG@10 of 0.499 [0.450, 0.543], followed by Line No-Header at 0.470 [0.414, 0.524], AST With-Header at 0.457 [0.405, 0.509], and AST No-Header at 0.345 [0.299, 0.390].

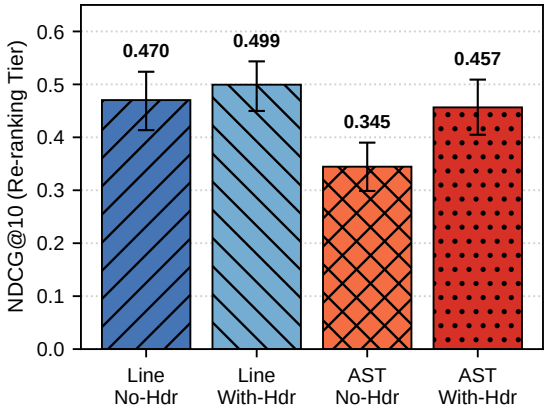


Fig. 2. NDCG@10 retrieval effectiveness across the four conditions at the two-stage re-ranking tier. Error bars represent 95% bootstrap confidence intervals across 25 JDs.

Figure 2 illustrates the re-ranking performance with 95% bootstrap confidence intervals. Figure 3 demonstrates the pronounced interaction effect: while context headers yield a modest non-significant improvement on line chunks ($\Delta = +0.029$), they recover $+0.112$ NDCG@10 on AST chunks ($p_{\text{Holm}} = 0.0031$).

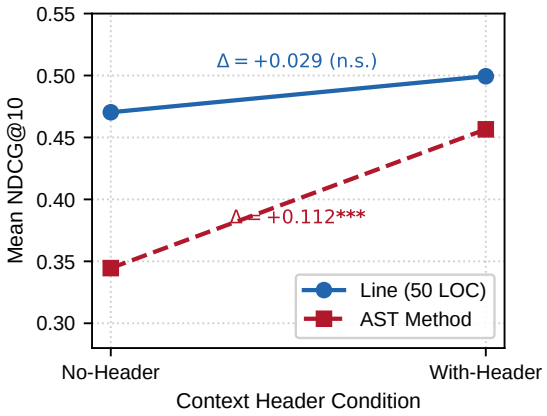


Fig. 3. Interaction plot demonstrating differential context header sensitivity between Line-based chunking and AST-based chunking.

TABLE III

RETRIEVAL EFFECTIVENESS ACROSS CHUNKING STRATEGIES, CONTEXT HEADER CONDITIONS, AND PIPELINE TIERS ($N = 25$ JOB DESCRIPTIONS).

Chunking Strategy	Header Condition	NDCG@10↑	NDCG@1↑	NDCG@3↑	NDCG@5↑	MRR@10↑	P@5↑	Recall@10↑	Judged@10
Panel A: Two-Stage Re-ranking Tier (Cross-Encoder BGE-Reranker-Large)									
Line (50 LOC)	No-Header	0.470 [0.414, 0.524]	0.740	0.703	0.587	0.933	0.656	0.246	**0.564**
Line (50 LOC)	With-Header	**0.499** [0.450, 0.543]	**0.800**	**0.736**	**0.630**	**0.980**	**0.704**	**0.262**	0.552
AST Method	No-Header	0.345 [0.299, 0.390]	0.460	0.477	0.399	0.846	0.560	0.220	0.536
AST Method	With-Header	0.457 [0.405, 0.509]	0.720	0.675	0.564	0.940	0.648	0.253	0.564
Panel B: Dense Bi-Encoder Tier (BGE-M3 Embeddings)									
Line (50 LOC)	No-Header	0.446 [0.377, 0.511]	0.600	0.593	0.512	0.903	0.640	0.265	**0.592**
Line (50 LOC)	With-Header	**0.450** [0.398, 0.502]	0.580	**0.600**	**0.539**	**0.913**	**0.704**	**0.273**	0.572
AST Method	No-Header	0.320 [0.265, 0.369]	0.480	0.426	0.378	0.807	0.568	0.217	0.548
AST Method	With-Header	0.408 [0.346, 0.463]	**0.660**	0.594	0.491	0.900	0.584	0.236	0.544
Panel C: Lexical Sparse Baseline Tier (BM25)[†]									
Line (50 LOC)	No-Header	0.104 [0.070, 0.141]	**0.140**	0.113	0.106	0.337	0.128	0.071	0.136
Line (50 LOC)	With-Header	0.129 [0.094, 0.165]	**0.140**	0.144	0.126	0.405	0.168	0.092	0.180
AST Method	No-Header	**0.187** [0.145, 0.230]	0.080	**0.171**	**0.173**	**0.409**	**0.288**	**0.148**	**0.304**
AST Method	With-Header	0.170 [0.122, 0.221]	0.080	0.151	0.167	0.395	0.248	0.120	0.244
Pre-registered Random Baselines									
Random Selection	Full Corpus (798 Chunks)	0.019	—	—	—	—	—	—	—
Random Selection	Judged Pool Candidates[‡]	0.592	—	—	—	—	—	—	—

Note: 95% bootstrap confidence intervals across 25 JDs (10,000 resamples, seed 42) shown in brackets below NDCG@10. In each panel, **bold** denotes the best performing configuration, and underline denotes the second-best configuration. Linear relevance gain $\mathrm{rel}/\log_2(i+1)$ is employed across all NDCG calculations.

[†] BM25 candidates were not included in the human judgment pool; unjudged items are treated as 0 (not relevant). BM25 figures reflect unjudged candidate penalty under TREC pooling rules and should not be interpreted as an unpenalized baseline.

[‡] Random selection within pool samples strictly from pre-judged items (rel $\in \{0, 1, 2\}$) and is non-comparable to open-retrieval systems facing unjudged candidates.

IV. HYPOTHESIS TESTING AND PAIRED COMPARISONS

Table IV presents the non-parametric Wilcoxon signed-rank tests for the confirmatory family and the exploratory contrast.

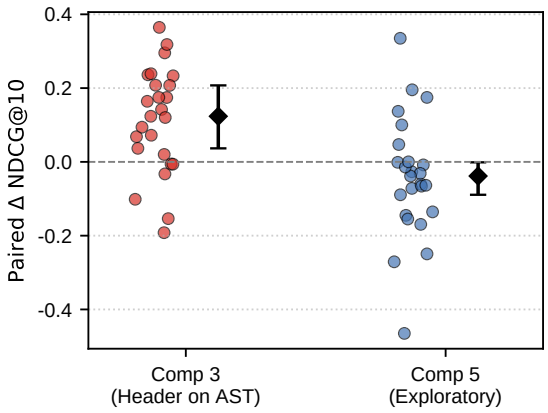


Fig. 4. Per-query paired difference (Δ NDCG@10) strip plot for Comp 3 (AST Header Effect) and Comp 5 (Exploratory AST vs. Line+Header). Black diamonds denote median and 95% bootstrap CI.

- **Comp 1 (Full Proposed vs. Baseline):** The overall difference between AST+Header and Line No-Header is $\Delta = -0.0138$ (95% CI $[-0.088, 0.058]$, $p_{\text{raw}} = 0.8532$, $p_{\text{Holm}} = 0.8532$). The pre-registered primary hypothesis is **not supported**.
- **Comp 2 (Pure Chunking):** In the absence of context headers, AST chunking underperforms Line chunking by $\Delta = -0.1258$ ($p_{\text{Holm}} = 0.0001$).
- **Comp 3 (Header on AST):** Adding structural context headers to AST chunks produces a large, statistically significant gain of $\Delta = +0.1120$ ($p_{\text{Holm}} = 0.0031$).
- **Comp 4 (Header on Line):** Context headers produce a minor, non-significant change on Line chunks ($\Delta = +0.0291$, $p_{\text{Holm}} = 0.7332$).
- **Comp 5 (Exploratory Contrast):** Line With-Header achieves an edge of $+0.0428$ over AST With-Header ($p_{\text{raw}} = 0.1355$, 95% CI $[-0.106, 0.021]$), which does not reach statistical significance.

Figure 4 displays the per-query paired differences for Comp 3 (19 wins, 6 losses) and Comp 5 (6 wins, 18 losses, 1 tie).

TABLE IV
HYPOTHESIS TESTING RESULTS FOR PRE-REGISTERED PRIMARY COMPARISONS AND EXPLORATORY CONTRAST (NDCG@10, RE-RANKING TIER).

ID	Hypothesis Description	Contrast ($A - B$)	Δ NDCG@10	95% Bootstrap CI	W/L/T	W	p_{raw}	p_{Holm}
Pre-registered Confirmatory Family (Step-down Holm-Bonferroni Correction, $\alpha = 0.05$)								
Comp 1	Full Proposed vs. Full Baseline	AST+Header vs. Line No-Header	-0.0138	[-0.088, 0.058]	12/13/0	155.0	0.8532	0.8532
Comp 2	Pure Chunking Effect	AST No-Header vs. Line No-Header	-0.1258	[-0.180, -0.076]	5/20/0	22.0	< 0.001	$< 0.001^{***}$
Comp 3	Context Header Effect on AST	AST+Header vs. AST No-Header	$+0.1120$	[0.056, 0.166]	19/6/0	46.0	0.0010	0.0031^{**}
Comp 4	Context Header Effect on Line	Line+Header vs. Line No-Header	$+0.0291$	[-0.017, 0.078]	15/10/0	128.0	0.3666	0.7332
Exploratory Analysis (Outside Pre-registered Family)								
Comp 5	AST+Header vs. Line+Header	AST+Header vs. Line+Header	-0.0428	[-0.106, 0.021]	6/18/1	97.0	0.1355	—

Note: $N = 25$ paired Job Descriptions. W: Wilcoxon signed-rank test statistic. Win/Loss/Tie (W/L/T) indicates the count of queries where System $A > B$, $A < B$, or $A = B$. Holm adjusted p-values (p_{Holm}) control family-wise error rate across Comps 1–4. Significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Comp 5 is an exploratory comparison outside the frozen registered confirmatory family and is not adjusted under Holm's step-down procedure.

V. SUBGROUP STRATIFICATION AND SENSITIVITY ANALYSES

Table V stratifies performance across Java ($N = 15$) and React/TypeScript ($N = 10$). In Java, AST+Header closely tracks Line+Header (0.5146 vs. 0.5358, $\Delta = -0.021$). In React/TypeScript, the gap widens to $\Delta = -0.075$ (0.3694 vs. 0.4448), driven by high line-fallback rates in React files.

Table VI tracks the composition of retrieved top-10 items. For AST No-Header, line-fallback chunks comprise 70.0% of top-10 results. Context headers increase pure AST retrieval to 54.0% at the re-ranking tier.

Table VII presents sensitivity audits. When removing all 122 disputed items (Panel A: Consensus Only), the AST header effect attenuates to $\Delta = +0.075$ and is **statistically non-significant** ($p = 0.0851$). Under judged-only evaluation (Panel C), Comp 1 remains non-significant ($\Delta = -0.043$, $p = 0.3666$).

TABLE V
RETRIEVAL PERFORMANCE (NDCG@10, RE-RANKING TIER) STRATIFIED BY TARGET PROGRAMMING LANGUAGE.

Configuration	Java ($N = 15$)	React/TS ($N = 10$)	Overall ($N = 25$)
Line No-Header	<u>0.525</u>	<u>0.388</u>	<u>0.470</u>
Line With-Header	**0.536**	**0.445**	**0.499**
AST No-Header	0.384	0.286	0.345
AST With-Header	0.515	0.369	0.457
Δ (AST+Hdr vs. Line+Hdr)†	-0.0212	-0.0754	-0.0428

Note: Stratification over 15 Java job descriptions and 10 React/TypeScript job descriptions at the two-stage re-ranking tier. **Bold** denotes the best performing configuration, and <u>underline</u> denotes the second-best within each language stratum. NDCG computed using linear gain $\mathrm{rel}/\log_2(i+1)$.
† Exploratory contrast (Comp 5). Negative values denote advantage for Line With-Header.

TABLE VI
AST FALLBACK COMPOSITION IN TOP-10 RETRIEVAL RESULTS ACROSS PIPELINE TIERS AND HEADER CONDITIONS.

Pipeline Tier	Header Condition	Total Top-10 (N)	AST Pure (%)	Line-Fallback (%)
BM25	No-Header	250	59 (23.6%)	191 (76.4%)
BM25	With-Header	250	130 (52.0%)	120 (48.0%)
Dense (BGE-M3)	No-Header	250	77 (30.8%)	173 (69.2%)
Dense (BGE-M3)	With-Header	250	166 (66.4%)	84 (33.6%)
Re-ranking (Cross-Encoder)	No-Header	250	75 (30.0%)	175 (70.0%)
Re-ranking (Cross-Encoder)	With-Header	250	135 (54.0%)	115 (46.0%)

Note: Evaluated across 25 JDs (250 total top-10 slots per configuration). Line-Fallback chunks occur when AST parsing fails or exceeds maximum LOC thresholds, defaulting to sliding-window chunks. Derived dynamically from `sensitivity_analysis_report.json`.

TABLE VII
SENSITIVITY ANALYSIS ACROSS GROUND TRUTH CURATION SUBSETS, SINGLE-ANNOTATOR QRELS, AND JUDGED-ONLY POOLS.

Evaluation Subset	Line+Hdr	Line No-Hdr	AST+Hdr	AST No-Hdr	AST Header Gain (Δ)	Statistical Decision
Panel A: Consensus-Only Ground Truth (122 Disagreements Excluded, $N = 481$ Pairs)						
Consensus Pairs	0.426	0.413	0.369	0.294	$+0.0749$ ($p = 0.0851$)	**Non-Significant** ($p \geq 0.05$)
Panel B: Single-Annotator Ground Truth Variants						
Annotator 1 Only	—	—	0.455	0.346	$+0.1094$ ($p < 0.001$)	Significant ($p < 0.001$)
Annotator 2 Only	—	—	0.475	0.378	$+0.0972$ ($p = 0.0012$)	Significant ($p < 0.01$)
Panel C: Judged-Only Candidate Evaluation (Unjudged Candidates Removed)						
Judged-Only Pool	0.615	0.590	0.546	0.454	$+0.0926$ ($p = 0.0034$)	Significant ($p < 0.01$)

Note: NDCG@10 at the two-stage re-ranking tier computed using linear gain $\mathrm{rel}/\log_2(i+1)$. In Panel A, removing 122 disputed items reduces the AST header effect to $\Delta = +0.075$ ($p = 0.0851$, statistically non-significant). In Panel C, removing unjudged candidates yields Comp 1 $\Delta = -0.043$ ($p = 0.3666$, non-significant) and exploratory Comp 5 $\Delta = -0.069$ ($p_{\mathrm{raw}} = 0.0318$). Data derived dynamically from `sensitivity_analysis_report.json`.

VI. CANDIDATE POOL AUDIT AND DATASET PROVENANCE

Table VIII provides a complete coverage audit. Top-3 judgment coverage was 100.0% across all Dense and Re-ranking configurations, while BM25 suffered unjudged rates of 65.3%–81.3% in top-3.

Table IX details pool deduplication (Panel A) and adjudication decisions (Panel B). Exactly 33 coordinate duplicates collapsed during deduplication. Adjudication resolved 122 disagreements into 73 agreements with Annotator 1, 47 with Annotator 2, and 2 independent resolutions.

Table X summarizes open-source licenses across the 40 mined evaluation repositories (19 MIT, 16 unspecified public GitHub, 4 Apache-2.0, 1 AGPL-3.0).

TABLE VIII
CANDIDATE POOL JUDGMENT COVERAGE AND UNJUDGED CANDIDATE EXPOSURE ACROSS 12 RETRIEVAL CONFIGURATIONS.

Configuration ID	Pipeline Tier	Top-3 Evaluated	Top-3 Unjudged	Judged@3 (%)	Top-10 Evaluated	Top-10 Unjudged	Judged@10 (%)
bm25_line_no_header	BM25†	75	61	18.7%	250	216	13.6%
bm25_line_with_header	BM25†	75	57	24.0%	250	205	18.0%
bm25_ast_no_header	BM25†	75	49	34.7%	250	174	30.4%
bm25_ast_with_header	BM25†	75	53	29.3%	250	189	24.4%
dense_line_no_header	Dense	75	0	100.0%	250	102	59.2%
dense_line_with_header	Dense	75	0	100.0%	250	107	57.2%
dense_ast_no_header	Dense	75	0	100.0%	250	113	54.8%
dense_ast_with_header	Dense	75	0	100.0%	250	114	54.4%
rerank_line_no_header	Rerank	75	0	100.0%	250	109	56.4%
rerank_line_with_header	Rerank	75	0	100.0%	250	112	55.2%
rerank_ast_no_header	Rerank	75	0	100.0%	250	116	53.6%
rerank_ast_with_header	Rerank	75	0	100.0%	250	109	56.4%

Note: Evaluated across 25 JDs (75 total top-3 slots, 250 total top-10 slots). Top-3 pool judgment was completely exhaustive (100.0%) across all 8 Dense and Re-rank configurations.

† BM25 configurations were excluded from candidate pooling; unjudged items (220 top-3 slots) are treated as 0 (not relevant). Derived dynamically from pool_coverage_report.csv.

TABLE IX
POOL DEDUPLICATION RECONCILIATION AND ADJUDICATION DECISION DISTRIBUTION.

Reconciliation / Adjudication Stage	Count	Proportion (%)
Panel A: Coordinate Deduplication & Pool Expansion		
Raw Recorded Top-3 Pool Instances	415	—
Unique Deduplicated Coordinate Pairs	382	92.0%
Pre-existing Overlap with Gold Standard	29	7.6%
New Unique Candidate Pairs Labeled	353	92.4%
Final Unified Ground Truth ($250 + 353$)	603	100.0%
Panel B: Discrepancy Adjudication Decisions ($N = 122$ Disagreements)		
Agreed with Annotator 1	73	59.8%
Agreed with Annotator 2	47	38.5%
Both Overruled (Independent Resolution)	2	1.6%
Structured Resolution Reasons (Distinct Strings)	70	—
— Explicit Prefix RESOLVED_BY_ADJUDICATION	29	23.8%
— Standard Evaluation Record Format	93	76.2%

Note: In Panel A, 33 duplicate coordinates arose from identical line boundaries shared between AST-fallback and Line-50 chunks. In Panel B, adjudication was performed by an independent lead adjudicator following structured templates. Derived from pool_reconciliation_report.txt and adjudication_summary.json.

TABLE X
DISTRIBUTION OF OPEN-SOURCE SOFTWARE LICENSES ACROSS THE 40 MINED EVALUATION REPOSITORIES.

License Category	Repositories (n)	Proportion (%)	Legal Classification
MIT License	19	47.5%	Permissive
Public GitHub (Not specified)	16	40.0%	Default Copyright
Apache-2.0	4	10.0%	Permissive
GNU AGPL-3.0	1	2.5%	Strong Copyleft
Total Evaluated Repositories	40	100.0%	20 Java / 20 React

Note: Tabulated from manifest_repos.csv. Mined snippets are distributed solely for peer-reviewed academic replication under scientific fair use.

APPENDIX A
CRYPTOGRAPHIC FILE PROVENANCE

Table XI records the SHA-256 digests and timestamps of all 14 data and result artifacts utilized to generate this report.

TABLE XI
CRYPTOGRAPHIC VERIFICATION AND TIMESTAMPS OF SOURCE FILES.

Relative File Path	SHA-256 (12)	Last Modified
dataset/ground_truth_annotator1.csv	c5e7f85b0d98	2026-10-10 10:05
dataset/ground_truth_annotator2.csv	45f799ef3c0d	2026-10-10 10:05
dataset/ground_truth_final.csv	2260c21debbc	2026-10-10 10:05
dataset/chunk_corpus.parquet	e7db3acebd90	2026-10-09 18:05
dataset/table_3_2_measured.json	13a0ffccf7e0	2026-10-11 12:37
dataset/table_3_2_report.txt	9b3de1d208b6	2026-10-11 12:37
dataset/benchmark_results/table_3_retrieval_benchmark.csv	45e8a9b36821	2026-10-11 12:35
dataset/benchmark_results/per_query_evaluation.csv	c9e263d7533d	2026-10-11 12:35
dataset/benchmark_results/statistical_tests_report.json	ae5d97f45a9c	2026-10-11 12:35
dataset/benchmark_results/sensitivity_analysis_report.json	fb9d228cb78a	2026-10-11 12:35
dataset/benchmark_results/pool_coverage_report.csv	04ca8918dee7	2026-10-10 13:47
dataset/benchmark_results/pool_reconciliation_report.txt	b560793baa37	2026-10-10 13:49
dataset/benchmark_results/adjudication_summary.json	9d46bcda1fce	2026-10-10 13:49
dataset/manifest_repos.csv	7bd2b90ca5a5	2026-10-10 11:50

APPENDIX B
REPRODUCTION INSTRUCTIONS

To reproduce all numbers, tables, figures, and compile this report from scratch, execute:

```
# 1. Run evaluation & tests
python scripts/evaluate_retrieval.py
python scripts/evaluate_sensitivity.py

# 2. Build report artifacts
python scripts/make_report.py

# 3. Verify numerical concordance
python -m pytest tests/test_report_numbers.py
```